

Chapter 5

Property, plant and equipment

Introduction

The main purpose of this chapter is to explain the requirements of international standard IAS16 *Property, Plant and Equipment*. This standard prescribes the accounting treatment of property, plant and equipment and deals with the following principal issues:

- (a) the criteria which must be satisfied in order that an item of property, plant and equipment should be recognised as an asset
- (b) the way in which items of property, plant and equipment should be measured in the financial statements, both at initial recognition and later
- (c) the way in which depreciation charges should be calculated and accounted for.

The accounting treatment prescribed by IAS16 applies to most property, plant and equipment but does not apply if another standard requires or permits a different accounting treatment. In particular, IAS16 does not apply to property, plant and equipment which is classified as held for sale. This is dealt with in accordance with IFRS5 (see Chapter 8).

This chapter of the book also considers four other international standards which may be relevant when accounting for property, plant and equipment. These are:

- (a) IFRS13 *Fair Value Measurement*
- (b) IAS23 *Borrowing Costs*
- (c) IAS20 *Accounting for Government Grants and Disclosure of Government Assistance*
- (d) IAS40 *Investment Property*.

Objectives

By the end of this chapter, the reader should be able to:

- define the term "property, plant and equipment" in accordance with IAS16
- state the criteria which must be satisfied for an item of property, plant and equipment to be recognised as an asset

- determine the cost of an item of property, plant and equipment
- distinguish between the cost model and the revaluation model for the measurement of property, plant and equipment after initial recognition
- calculate and account for depreciation charges in accordance with IAS16
- list the main disclosures required by IAS16
- explain the requirements of IAS23 in relation to borrowing costs incurred for the acquisition, construction or production of a qualifying asset
- explain the requirements of IAS20 in relation to government grants
- define the term "investment property" and explain the requirements of IAS40 in relation to investment properties
- define the term "fair value" and outline the guidance provided by IFRS13 with regard to the measurement of fair value.

Definition of property, plant and equipment

IAS16 defines property, plant and equipment as "*tangible items that:*

- (a) *are held for use in the production or supply of goods or services, for rental to others, or for administrative purposes; and*
- (b) *are expected to be used during more than one period."*

This definition covers the majority of the tangible non-current assets that might normally be held by a business entity.

Definition of carrying amount

There are several other definitions in IAS16, most of which are covered in later sections of this chapter. But one definition which must be understood from the outset is the definition of an asset's carrying amount. The term "carrying amount" is used throughout the international standards to refer to the amount at which an item is "carried" or shown in the financial statements. In the case of property, plant and equipment, IAS16 defines carrying amount as "*the amount at which an asset is recognised after deducting any accumulated depreciation and accumulated impairment losses*". This amount could also be referred to as the asset's written down value, although this term is not used by the international standards. Impairment losses are dealt with in Chapter 7 of this book.

Recognition of property, plant and equipment

IAS16 states that an item of property, plant and equipment should be recognised as an asset if and only if:

- (a) it is probable that future economic benefits associated with the item will flow to the entity concerned, and
- (b) the cost of the item can be measured reliably[†].

Assessing the probability of future economic benefits will require the exercise of judgement but measuring the cost of an item will usually be straightforward. IAS16 also makes the following points with regard to the recognition of property, plant and equipment:

- (a) An "item" generally consists of an individual piece of property, plant and equipment. However, it may sometimes be appropriate to group relatively insignificant items together (e.g. small tools) and apply the recognition criteria to the group.
- (b) Spare parts are recognised as property, plant and equipment if they meet the definition given earlier in this chapter and satisfy the recognition criteria. However, small parts may be treated as inventory (see Chapter 10) and accounted for as an expense when they are used in the repair and maintenance of property, plant and equipment.
- (c) Items of property, plant and equipment acquired for safety or environmental reasons are recognised as assets because they enable the entity to derive future economic benefits from its other property, plant and equipment. For instance, a manufacturer might be obliged by law to instal certain safety equipment and would be unable to operate any production machinery at all if this safety equipment were not installed.

[†] *These criteria are in accordance with the recognition criteria for assets laid down in the 2010 Conceptual Framework and have not yet been amended to reflect the revised recognition criteria established in the 2018 Conceptual Framework (see Chapter 2).*

Subsequent costs

Routine servicing, repair and maintenance costs incurred after the initial recognition of an item of property, plant and equipment as an asset are not capital expenditure. These costs usually consist of labour costs and the cost of small parts (see above) and are accounted for as an expense in the period to which they relate.

However, it may be necessary to replace major parts of some items of property, plant and equipment at regular intervals (e.g. the interior fittings of an aircraft). IAS16 requires that the cost of major replacements should be treated as capital expenditure and recognised as an addition to the carrying amount of the asset concerned. The carrying amount of the part that has been replaced should simultaneously be derecognised (see below).

A similar procedure applies to the cost of major inspections. If an item of property, plant and equipment which has been recognised as an asset needs regular major inspections, the cost of each inspection is added to the carrying amount of the asset when it occurs. Any remaining carrying amount of the cost of the previous inspection is derecognised.

Derecognition

An item of property, plant and equipment previously recognised as an asset should be derecognised (i.e. removed from the statement of financial position) when:

- (a) it is disposed of, or
- (b) no future economic benefits are expected from its use or from its disposal.

IAS16 states that any gain or loss arising on the derecognition of an item of property, plant and equipment should be "included in profit or loss" when the item is derecognised. This means that the gain or loss should be included in the calculation of the entity's profit or loss for the period in which derecognition occurs.

The gain or loss arising on derecognition is equal to the difference between the disposal proceeds (if any) and the asset's carrying amount at the time of derecognition.

EXAMPLE 1

- (a) On 1 September 2019, a company paid £80,000 to replace the wall lining of one of its furnaces. The furnace had been acquired several years previously and its carrying amount on 1 September 2019 (before accounting for the replacement of the lining) was £320,000. Of this amount, £10,000 related to the original wall lining.
- (b) On 1 September 2019, a company paid £250,000 for a major inspection of one of its aircraft. It is a legal requirement that such an inspection is carried out at least once every three years. The previous inspection took place in March 2017 at a cost of £210,000. The carrying amount of the aircraft on 1 September 2019 (before accounting for the new inspection) was £1,200,000. Of this amount, £70,000 related to the previous inspection.

Explain how each of these transactions should be accounted for in accordance with the requirements of IAS16.

Solution

- (a) The cost of the replacement wall lining should be recognised as an asset and the carrying amount of the original lining should be derecognised. The carrying amount of the furnace becomes £390,000 ($£320,000 + £80,000 - £10,000$). The gain or loss arising on the disposal of the old lining is included in the calculation of the company's profit or loss for the accounting period in which derecognition occurs. This gain or loss is calculated as the difference between the £10,000 carrying amount of the old lining and any amount which is received on disposal.
 - (b) The cost of the new inspection should be recognised as an asset and the carrying amount of the previous inspection should be derecognised. The carrying amount of the aircraft becomes £1,380,000 ($£1,200,000 + £250,000 - £70,000$). The £70,000 carrying amount of the previous inspection is written off as an expense.
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Initial measurement of property, plant and equipment

On initial recognition, IAS16 requires that items of property, plant and equipment should be measured at cost. The cost of an item of property, plant and equipment comprises:

- (a) the purchase price of the item, including import duties and non-refundable purchase taxes, less trade discounts or rebates
- (b) costs that are directly attributable to bringing the item to the location and condition necessary for it to be operated as intended, including:
 - (i) labour costs arising directly from the construction or acquisition of the item
 - (ii) site preparation costs
 - (iii) initial delivery and handling costs
 - (iv) installation, assembly and testing costs
 - (v) professional fees
- (c) the estimated costs of dismantling and removing the item and restoring the site on which the item is located, as long as the obligation to meet these costs is incurred when the item is acquired or used (see Chapter 12).

In general, administrative costs and other general overhead expenses are not part of the cost of an item of property, plant and equipment.

EXAMPLE 2

On 31 July 2019, a company which prepares financial statements to 31 March each year bought a machine for £648,000. This amount was made up as follows:

	£
Manufacturer's list price	500,000
Less: Trade discount	30,000
	470,000
Delivery charge	4,300
Installation and testing charges	24,500
Small spare parts	5,200
Servicing contract for the year to 31 July 2020	36,000
	540,000
Value added tax at 20%	108,000
	648,000

The company is VAT-registered and reclaims VAT charged to it by its suppliers. Calculate the cost of the machine in accordance with the requirements of IAS16.

Solution

The machine's cost is £498,800 (£470,000 + £4,300 + £24,500). As regards the other amounts listed above:

- (a) The small spare parts will probably be treated as inventory and should be accounted for as an expense when used.
- (b) Eight-twelfths of the cost of the servicing contract (£24,000) should be accounted for as an expense in the year to 31 March 2020. The remaining four-twelfths (£12,000) is a prepayment of an expense for the year to 31 March 2021.
- (c) VAT can be ignored since it is refunded to the company.

Self-constructed assets

The cost of a self-constructed asset (e.g. an office building constructed by a building firm for its own occupation) is determined in the same way as the cost of any other item of property, plant and equipment. If the entity sells similar assets to customers in the normal course of trade, the cost of the self-constructed asset will usually be the same as the cost of constructing an asset for sale. This should exclude any profit element and is calculated in accordance with IAS2 *Inventories* (see Chapter 10).

The accounting treatment of interest and other borrowing costs incurred in connection with the acquisition, construction or production of an asset is prescribed by international standard IAS23 *Borrowing Costs* (see later in this chapter).

Subsequent measurement of property, plant and equipment

IAS16 allows two different ways of measuring property, plant and equipment subsequent to its initial recognition as an asset. These are the "cost model" and the "revaluation model". A brief description of each model is as follows:

- (a) **The cost model.** After initial recognition, items of property, plant and equipment are carried at cost less any accumulated depreciation and less any accumulated impairment losses. Impairment losses are dealt with in Chapter 7 of this book.
- (b) **The revaluation model.** After initial recognition, items of property, plant and equipment are carried at a revalued amount. The revalued amount of an item consists of its fair value at the date of revaluation, less any subsequent accumulated depreciation and less any subsequent accumulated impairment losses. Note that:
 - (i) IAS16 defines fair value as "*the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date*". This definition is derived from international standard IFRS13 *Fair Value Measurement*, which sets out a framework for measuring the fair value of an asset or liability. A brief summary of IFRS13 is provided in the appendix at the end of this chapter.

- (ii) The revaluation model cannot be applied to an item of property, plant and equipment unless its fair value can be measured reliably.
- (iii) If the revaluation model is applied to an item of property, plant and equipment, then it must be applied to the entire class to which the item belongs. A class of property, plant and equipment is a grouping of items of a similar nature. Typical classes might include land, buildings, machinery, motor vehicles etc.
- (iv) Revaluations should be made with sufficient regularity to ensure that the carrying amount of an item is not materially different from its fair value at the end of the reporting period. All of the items within a class should be revalued simultaneously. This avoids selective revaluations and also avoids the reporting of amounts that are a mixture of valuations made on different dates.

Accounting for revaluation gains and losses

If the carrying amount of an item of property, plant and equipment is *increased* as a result of a revaluation, the increase must normally be credited to a revaluation reserve and shown as "other comprehensive income" in the entity's statement of comprehensive income. This accounting treatment ensures that unrealised revaluation gains (i.e. gains that have not been turned into cash) are excluded from profit and so are not available for payment as a dividend. If the carrying amount of an item of property, plant and equipment is *decreased* as a result of a revaluation, the decrease must normally be recognised as an expense when calculating the entity's profit or loss. But note that:

- (a) a revaluation increase must be recognised as income when calculating the entity's profit or loss to the extent that it reverses any revaluation decrease in respect of the same item that was previously recognised as an expense
- (b) a revaluation decrease must be debited to the revaluation reserve and shown (as a negative figure) in other comprehensive income to the extent of any credit balance previously existing in the revaluation reserve in respect of that same item.

Disposal of a revalued item of property, plant and equipment

When an item of property, plant and equipment is disposed of, any revaluation gain which is included in the revaluation reserve in respect of that item may be transferred to retained earnings. This transfer takes account of the fact that a previously unrealised gain has now been realised. The transfer is recorded in the statement of changes in equity and does not affect the statement of comprehensive income.

Note that a revaluation gain is an example of an item of other comprehensive income that will not be reclassified to profit or loss in a subsequent period (see Chapter 3). The fact that the transfer described above does not occur in the statement of comprehensive income means that the gain is excluded from profit both initially (when the asset is revalued) and subsequently (when it is disposed of).

EXAMPLE 3

- (a) Company X prepares financial statements to 31 May each year. On 31 May 2019, the company acquired land for £400,000. This land was revalued at £450,000 on 31 May 2020 and at £375,000 on 31 May 2021.
- (b) Company Y prepares financial statements to 30 June each year. On 30 June 2019, the company acquired land for £600,000. This land was revalued at £540,000 on 30 June 2020 and at £620,000 on 30 June 2021.

Assuming that both companies use the revaluation model, explain how each revaluation should be dealt with in the financial statements. Ignore depreciation.

Solution

- (a) The £50,000 revaluation increase on 31 May 2020 should be credited to a revaluation reserve and shown as other comprehensive income in the company's statement of comprehensive income.
- £50,000 of the £75,000 decrease on 31 May 2021 should be debited to revaluation reserve and shown (as a negative figure) in other comprehensive income in the statement of comprehensive income. The remaining £25,000 should be recognised as an expense when calculating the company's profit or loss for the year.
- (b) The £60,000 revaluation decrease on 30 June 2020 should be recognised as an expense when calculating the company's profit or loss for the year.
- £60,000 of the £80,000 increase on 30 June 2021 should be recognised as income when calculating the company's profit or loss for the year. The remaining £20,000 should be credited to a revaluation reserve and shown as other comprehensive income in the statement of comprehensive income.

Depreciation

IAS16 defines depreciation as "*the systematic allocation of the depreciable amount of an asset over its useful life*". This definition makes it clear that the purpose of depreciation is simply to allocate an expense between accounting periods. The depreciation process makes no attempt to show assets at their current values. Nor does it guarantee that there will be funds available to replace those assets when they come to the end of their useful lives. Depreciation charges reduce profits but have no direct effect on an entity's cash resources and do not ensure that cash is "saved up" to buy replacement assets.

IAS16 also defines a number of key terms which are used in the section of the standard which deals with depreciation. These are as follows:

- (a) Depreciable amount is "*the cost of an asset, or other amount substituted for cost, less its residual value*".

- (b) The residual value of an asset is *"the estimated amount that an entity would currently obtain from disposal of the asset, after deducting the estimated costs of disposal, if the asset were already of the age and in the condition expected at the end of its useful life"*.
- (c) Useful life is *"the period over which an asset is expected to be available for use by an entity ... or the number of production or similar units expected to be obtained from the asset by an entity"*.

The definition of depreciable amount makes it clear that depreciation does not cease if an asset is revalued. All that happens is that a valuation is substituted for the asset's carrying amount, resulting in a revaluation gain or loss. Depreciable amount is now equal to the amount of the valuation less the asset's residual value. Subsequent depreciation charges will be based upon this revised depreciable amount and will continue for the remainder of the asset's useful life.

If an item of property, plant and equipment is revalued upwards and a revaluation gain in relation to that item is credited to revaluation reserve, subsequent depreciation charges will exceed depreciation based on the item's original cost. In these circumstances, the amount of excess depreciation in each accounting period may be transferred from revaluation reserve to retained earnings. This transfer takes place in the statement of changes in equity and does not affect the statement of comprehensive income.

IAS16 requirements with regard to depreciation

IAS16 establishes a number of general requirements with regard to the depreciation of property, plant and equipment and then deals with depreciation methods. The main general requirements are as follows:

- (a) Each part of an item of property, plant and equipment with a cost that is significant in relation to the item's total cost should be depreciated separately. However, parts of an item may be grouped for depreciation purposes if they have the same useful life and if the same depreciation method is applicable to each part. An example of the need to depreciate the parts of an item separately arises in the case of an aircraft, where the airframe and the engines may have very different useful lives.
- (b) The depreciation charge for an accounting period should normally be recognised as an expense when calculating profit or loss. However, if an item of property, plant and equipment is used to produce other assets, then the depreciation of that item should be included in the cost of those other assets. For instance, the depreciation of factory machines may be included in the cost of inventories (see Chapter 10).
- (c) The residual value and useful life of property, plant and equipment should be reviewed at least at the end of each financial year. If expectations differ from previous estimates, the revised estimates should be accounted for prospectively, as required by IAS8 (see Chapter 4). The asset's remaining depreciable amount should be amended to reflect any change in residual value and then this amount should be allocated as depreciation over the remainder of the asset's expected useful life.

- (d) A number of factors should be taken into account when judging the useful life of an asset. These include expected usage of the asset, expected physical wear and tear and the likelihood of technical or commercial obsolescence. Useful life is defined in terms of the usefulness of the asset to the entity concerned and this may be shorter than the asset's full economic life.
- (e) If the residual value of an asset is greater than or equal to its carrying amount, then the asset's depreciable amount is zero. This means that depreciation charges will also be zero until such time as the asset's residual value becomes less than its carrying amount. This may or may not happen.
- (f) Depreciation should begin when an asset is available for use and then continue until the asset is derecognised or classified as held for sale (see Chapter 8). Depreciation should not cease simply because an asset has become idle. However, some methods of depreciation (see below) are based on an asset's usage and would give a zero depreciation charge whilst the asset is not being used.
- (g) Land and buildings should be dealt with separately from one another for depreciation purposes. In most cases, land has an unlimited useful life and is not depreciated for that reason. But an exception arises in the case of quarries, mines etc. which generally have finite useful lives and so should be depreciated.

Depreciation methods

IAS16 states that a variety of depreciation methods may be used to allocate the depreciable amount of an item of property, plant and equipment over its useful life. These include (but are not limited to) the following well-known methods:

- (a) **Straight-line method.** The asset's depreciable amount is spread evenly over its useful life, so giving a constant depreciation charge in each accounting period.
- (b) **Diminishing balance (or reducing balance) method.** The depreciation charge for an accounting period is calculated by applying a constant percentage to the carrying amount of the asset brought forward from the previous period. This method gives a decreasing depreciation charge in each successive period.
- (c) **Units of production method.** This method is typically applied to machines used in a factory. The useful life of the asset is measured in units of production rather than years and the depreciation charge for an accounting period is calculated in accordance with the number of units of production achieved in that period.

This is very similar to the machine hours method, where the useful life of a machine is measured in working hours and the depreciation charge for an accounting period is calculated in accordance with the number of hours for which the machine has been used during the period.

In principle, the chosen method should be the one which most closely matches the usage pattern of the asset concerned. This is what IAS16 requires.

The depreciation method applied to an asset should be reviewed at least at the end of each financial year and changed if there has been a change in usage pattern. Such a change should be accounted for as a change in an accounting estimate in accordance with the requirements of IAS8 (see Chapter 4). This standard requires the new depreciation method to be applied prospectively over the remainder of the asset's useful life.

EXAMPLE 4

On 1 January 2019, a company which prepares financial statements to 31 December each year buys a machine at a cost of £46,300. The machine's useful life is estimated at four years with a residual value of £6,000. The machine is expected to achieve 50,000 units of production over its useful life, as follows:

	2019	2020	2021	2022
Number of units	10,000	20,000	15,000	5,000

Calculate depreciation charges for each of these four years using:

- the straight-line method
- the diminishing balance method (at a rate of 40%)
- the units of production method.

Solution

- The machine's depreciable amount is £40,300. Allocating this evenly over the four years gives an annual depreciation charge of £10,075.

(b) Year	Carrying amount b/f £	Depreciation at 40% £	Carrying amount c/f £
2019	46,300	18,520	27,780
2020	27,780	11,112	16,668
2021	16,668	6,667	10,001
2022	10,001	4,001	6,000
		<u>40,300</u>	

- Depreciation per unit is 80.6p (£40,300 ÷ 50,000). Depreciation charges are:

Year	Units of production £	Depreciation at 80.6p £
2019	10,000	8,060
2020	20,000	16,120
2021	15,000	12,090
2022	5,000	4,030
		<u>40,300</u>

Disclosure requirements

The disclosure requirements of IAS16 are very extensive. The main disclosures required are as follows:

- (a) for each class of property, plant and equipment:
 - (i) the measurement bases used
 - (ii) the depreciation methods used
 - (iii) the useful lives or depreciation rates used
 - (iv) the gross carrying amount and accumulated depreciation at the beginning and end of the accounting period
 - (v) a reconciliation of the carrying amount at the beginning and end of the period, showing additions, disposals, revaluation increases and decreases, depreciation, impairment losses and any other movements
- (b) the amount of any property, plant and equipment pledged as security for liabilities
- (c) the amount of any contractual commitments for the acquisition of property, plant and equipment
- (d) for items of property, plant and equipment stated at revalued amounts (in addition to the disclosures required by IFRS13 *Fair Value Measurement*):
 - (i) the date of the revaluation
 - (ii) whether or not an independent valuer was involved
 - (iii) for each revalued class of property, plant and equipment, the carrying amount that would have been recognised if the cost model had been used.

IAS16 also encourages (but does not require) entities to disclose additional information which may be relevant to the needs of users. This additional information might include:

- (e) the carrying amount of temporarily idle property, plant and equipment
- (f) the gross carrying amount of any fully-depreciated property, plant and equipment that is still in use
- (g) the fair value of property, plant and equipment to which the cost model has been applied, if this is materially different from the carrying amount.

Borrowing costs

The objective of IAS23 *Borrowing Costs* is to prescribe the accounting treatment of interest and other borrowing costs, especially those which are incurred in connection with the acquisition of assets such as property, plant and equipment.

The "core principle" of IAS23 is that borrowing costs that are directly attributable to the acquisition, construction or production of a "qualifying asset" should be capitalised as part of the cost of that asset. Other borrowing costs should be recognised as an expense. The standard provides the following definitions:

- (a) Borrowing costs are *"interest and other costs that an entity incurs in connection with the borrowing of funds"*. Borrowing costs include:
 - (i) costs accounted for by the "effective interest method" (see Chapter 11) such as:
 - interest on bank overdrafts and on short-term and long-term borrowings
 - discounts and premiums related to borrowings
 - transaction costs incurred in connection with the arrangement of borrowings
 - (ii) interest arising in respect of lease liabilities (see Chapter 9).
- (b) A qualifying asset is *"an asset that necessarily takes a substantial period of time to get ready for its intended use or sale"*. Depending on the circumstances, any of the following may be qualifying assets:
 - inventories
 - manufacturing plants and power generation facilities
 - intangible assets
 - investment properties.

Inventories produced over a short period of time are not qualifying assets. Nor are assets which are ready for their intended use or sale as soon as they are acquired.

Borrowing costs eligible for capitalisation

The borrowing costs that are directly attributable to the acquisition, construction or production of a qualifying asset are those borrowing costs which would have been avoided if the expenditure on the asset had not occurred. Identifying these costs is straightforward if funds are borrowed specifically for this purpose. In these circumstances, the borrowing costs eligible for capitalisation are the actual borrowing costs incurred on the borrowed funds during the "period of capitalisation" (see below).

If a qualifying asset is funded out of a pool of general borrowings, the borrowing costs that may be capitalised are calculated by applying a "capitalisation rate" to the expenditure on the asset. The capitalisation rate is the weighted average of the borrowing costs that are applicable to the general borrowings outstanding during the period of capitalisation.

Period of capitalisation

The capitalisation of borrowing costs incurred in relation to a qualifying asset should begin as soon as *all* of the following conditions are satisfied:

- (a) expenditure is being incurred on the asset, and
- (b) borrowing costs are being incurred, and
- (c) activities are in progress that are necessary to prepare the asset for its intended use.

Capitalisation of borrowing costs ceases when substantially all of the activities necessary to prepare the asset for its intended use are complete.

Disclosure requirements

An entity is required to disclose:

- (a) the amount of borrowing costs capitalised during the accounting period
- (b) the capitalisation rate used to calculate the amount of general borrowing costs that are eligible for capitalisation.

EXAMPLE 5

A company which prepares financial statements to 30 June each year has the following general borrowings outstanding throughout the year to 30 June 2020:

	<i>£000</i>
7.5% Bank loan	800
9% Bank loan	500
8.5% Loan stock	1,200
	2,500

On 1 October 2019, the company began construction of a qualifying asset and incurred expenditure of £300,000. A further £240,000 was spent on 1 February 2020. Both of these amounts were financed out of general borrowings. Construction of the asset was still underway at 30 June 2020.

Calculate the amount of borrowing costs that should be capitalised as part of the cost of the qualifying asset during the year to 30 June 2020.

Solution

The total borrowing costs incurred in relation to general borrowings during the year to 30 June 2020 were $(£800,000 \times 7.5\%) + (£500,000 \times 9\%) + (£1,200,000 \times 8.5\%) = £207,000$.

On total borrowings of £2.5 million, this represents a weighted average rate of 8.28%. The borrowing costs which should be capitalised are therefore $(£300,000 \times 8.28\% \times 9/12) + (£240,000 \times 8.28\% \times 5/12) = £26,910$.

Government grants

The main objective of international standard IAS20 *Accounting for Government Grants and Disclosure of Government Assistance* is to prescribe the accounting treatment of government grants. IAS20 defines government grants as "*assistance by government in the form of transfers of resources to an entity in return for past or future compliance with certain conditions relating to the operating activities of the entity*". Government grants are of two main types:

- (a) **Grants related to assets.** These are government grants which are made on condition that the entity should purchase, construct or otherwise acquire long-term assets such as property, plant and equipment. The purpose of such a grant is to help the entity to pay for the asset or assets concerned.
- (b) **Grants related to income.** These are government grants other than grants related to assets. The usual purpose of a grant related to income is to help the entity to pay some of its operating expenses.

The first requirement of IAS20 is that government grants should not be recognised in the financial statements until there is reasonable assurance that the entity will comply with the relevant conditions and the grant will in fact be received. Once this assurance exists, government grants should be "*recognised in profit or loss on a systematic basis over the periods in which the entity recognises as expenses the related costs for which the grants are intended to compensate*".

This means that a grant for the acquisition of an asset should be recognised as income when calculating the entity's profit or loss for the accounting periods in which depreciation is charged on that asset. In each period, the proportion of the grant which is recognised as income should be the same as the proportion of the asset's depreciable amount which is charged as depreciation in that period. IAS20 permits two ways of achieving this effect:

- (a) The grant may be credited to a deferred income account and then systematically transferred to the statement of comprehensive income over the useful life of the asset.
- (b) The grant may be deducted from the carrying amount of the asset. This will result in reduced depreciation charges over the asset's useful life.

The two methods are equivalent but give a different presentation in the statement of comprehensive income and in the statement of financial position.

EXAMPLE 6

On 1 April 2019, a company which prepares financial statements to 31 March acquires an item of property, plant and equipment and receives a government grant of 40% towards the item's cost. The item costs £250,000 and has a useful life of four years. Its residual value is £60,000.

- (a) Calculate the amount of the grant that should be recognised as income for each of the four years to 31 March 2023 if the asset is depreciated:
- on the straight-line basis
 - on the diminishing balance basis (at a rate of 30%).
- (b) Assuming that the straight-line basis is used, explain how the grant will be dealt with in the financial statements if:
- the grant is treated as deferred income
 - the grant is deducted from the cost of the asset.

Solution

- (a) The grant is £100,000 and the asset's depreciable amount is £190,000. The amount of the grant that should be recognised as income each year is calculated as follows:

(i) **Straight-line basis**

	<i>Depreciation charge</i>	<i>Grant recognised as income</i>
	£	£
year to 31 March 2020	47,500	25,000
year to 31 March 2021	47,500	25,000
year to 31 March 2022	47,500	25,000
year to 31 March 2023	47,500	25,000
	<u>190,000</u>	<u>100,000</u>

Since the asset is depreciated on the straight-line basis, the grant is recognised as income on the straight-line basis.

(ii) **Diminishing balance basis**

	<i>Carrying amount b/f</i>	<i>Depreciation charge</i>	<i>Grant recognised as income</i>
	£	£	£
year to 31 March 2020	250,000	75,000	39,474
year to 31 March 2021	175,000	52,500	27,632
year to 31 March 2022	122,500	36,750	19,342
year to 31 March 2023	85,750	25,750	13,552
		<u>190,000</u>	<u>100,000</u>

Each year's depreciation charge is 30% of the carrying amount brought forward, except that the charge for 2023 has been rounded to ensure that the sum of the depreciation charges is exactly equal to the depreciable amount.

In 2020, the depreciation charge of £75,000 is approximately 39.474% of the depreciable amount of £190,000, so 39.474% of the grant is recognised as income in that year. A similar calculation is used in the remaining three years to calculate the amount of the grant recognised as income.

- (b) (i) The grant of £100,000 is credited to a deferred income account and transferred in equal instalments to the statement of comprehensive income. The figures appearing in the statement of comprehensive income and in the statement of financial position are as follows:

	2020 £	2021 £	2022 £	2023 £
Statement of comprehensive income:				
Depreciation charge	47,500	47,500	47,500	47,500
Grant income	25,000	25,000	25,000	25,000
Statement of financial position:				
Asset carrying amount	202,500	155,000	107,500	60,000
Deferred income a/c	75,000	50,000	25,000	0

- (ii) In this case, the cost of the asset (net of the grant) is £150,000. Residual value is £60,000, so depreciable amount is £90,000. This is written off as an expense on the straight-line basis over the four years, as follows:

	2020 £	2021 £	2022 £	2023 £
Statement of comprehensive income:				
Depreciation charge	22,500	22,500	22,500	22,500
Statement of financial position:				
Asset carrying amount	127,500	105,000	82,500	60,000

Disclosure requirements of IAS20

IAS20 requires that entities should disclose:

- the accounting policy adopted in relation to government grants, including the method of presentation adopted in the financial statements
- the nature and extent of government grants recognised in the financial statements.
- any unfulfilled conditions attaching to government grants that have been recognised.

Investment property

There is a presumption in IAS16 that property, plant and equipment is acquired for use in an entity's operations and is generally used up or "consumed" over the period of its useful life. Depreciation charges are made so as to match the cost of consuming an asset against the revenues which it helps to generate. But property which is acquired as an investment rather than for use is not consumed in the entity's operations and does not have a useful life. This means that the accounting treatment required by IAS16 is generally inappropriate for such property. The objective of IAS40 *Investment Property* is to offer a more suitable accounting treatment for property held as an investment.

Definition of investment property

IAS40 defines investment property as "*property (land or a building - or part of a building - or both) held ... to earn rentals or for capital appreciation or both, rather than for:*

- (a) *use in the production or supply of goods or services or for administrative purposes; or*
- (b) *sale in the ordinary course of business".*

Examples of investment property given in IAS40 include the following:

- (a) land held for long-term capital appreciation
- (b) land held for a currently undetermined future use
- (c) a building that is leased out under one or more operating leases (see Chapter 9)
- (d) a building that is vacant but is held to be leased out under operating leases
- (e) property that is being constructed or developed for future use as investment property.

The standard also provides examples of items that are *not* investment property and are not within the scope of IAS40. These include:

- (a) property intended for sale in the ordinary course of business; the applicable standard is IAS2 *Inventories* (see Chapter 10)
- (b) owner-occupied property (including property held for future use as owner-occupied property); the applicable standard is IAS16 *Property, Plant and Equipment*
- (c) property leased to another entity under a finance lease; the applicable standard is IFRS16 *Leases* (see Chapter 9).

Measurement of investment property

Investment property should be measured initially at cost. This includes transaction costs such as legal fees and property transfer taxes. As regards the measurement of investment property subsequent to its initial recognition, IAS40 requires an entity to choose between two models. Whichever model is adopted as the entity's accounting policy must normally be applied to all of the entity's investment property. The two models are as follows.

- (a) **The fair value model.** Investment property is measured at its fair value at the end of each reporting period. As stated earlier in this chapter, IFRS13 defines fair value as *"the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date"*.

A gain or loss arising from a change in the fair value of investment property must be recognised in the calculation of profit or loss for the period in which it arises. This is different from the IAS16 treatment of revaluation gains and losses, since IAS16 requires revaluation gains to be excluded from profit (see earlier in this chapter).

- (b) **The cost model.** This is the cost model as defined in IAS16. Investment property is measured at cost, less any accumulated depreciation and less any accumulated impairment losses. The *Basis for Conclusions* for IAS40 suggests that this model might be used in countries with less-developed property markets and valuation professions.

An entity which generally adopts the fair value model in relation to investment property must nonetheless apply the cost model to a specific investment property if the fair value of that property cannot be determined reliably.

Disclosure requirements

The general disclosure requirements of IAS40 (which are in addition to the disclosures required by IFRS16 *Leases*) are as follows:

- (a) whether the entity has adopted the fair value model or the cost model
- (b) the extent to which fair values have been determined by an independent valuer who holds a recognised and relevant professional qualification
- (c) the amounts of any rental income from investment property and related expenses that have been recognised when calculating the entity's profit or loss for the period.

If the fair value model is adopted, the entity must also provide a reconciliation between the carrying amounts of investment property at the beginning and end of the period, showing additions, disposals, gains or losses from fair value adjustments and any other movements.

If the cost model is adopted, the entity must disclose similar information to that required by IAS16 for property, plant and equipment measured using the cost model. The entity must also disclose the fair value of its investment property (unless this cannot be determined reliably). This means that fair values will normally have to be determined whether or not the entity has actually adopted the fair value model.

Summary

- ▶ Property, plant and equipment consists of tangible non-current assets that are held for use in the production or supply of goods or services, or for rental to others or for administrative purposes.
- ▶ Property, plant and equipment is recognised as an asset when it is probable that it will yield future economic benefits and its cost can be measured reliably. It should be derecognised on disposal or when no future economic benefits are expected from it.
- ▶ Property, plant and equipment is measured initially at cost. Subsequently, an entity may apply either the cost model or the revaluation model.
- ▶ In general, revaluation gains are credited to a revaluation reserve and recognised as other comprehensive income in the statement of comprehensive income. Revaluation losses are recognised as an expense when calculating profit or loss for the period.
- ▶ The purpose of depreciation is to allocate the depreciable amount of an asset over its useful life. Residual values and lengths of useful lives should be reviewed at regular intervals. A depreciation method should be chosen that best matches the usage pattern of the asset concerned.
- ▶ Borrowing costs that are directly attributable to the acquisition, construction or production of a qualifying asset must be capitalised as part of the cost of that asset. Other borrowing costs must be recognised as an expense.
- ▶ Government grants relating to the acquisition of an asset are recognised as income over the useful life of that asset. The grant may be credited to a deferred income account and then systematically transferred to the statement of comprehensive income. Alternatively, the grant may be deducted from the carrying amount of the asset, resulting in reduced depreciation charges over the asset's useful life.
- ▶ Investment property consists of land or buildings held for the purpose of earning rental income or for capital appreciation or both.
- ▶ Investment property is measured initially at cost. Subsequently, an entity may apply either the fair value model or the cost model.

Appendix

IFRS13 Fair Value Measurement

The objectives of IFRS13 are to define "fair value", to set out a framework for measuring fair value, and to specify the required disclosures about fair value measurements.

Definition of fair value

IFRS13 defines fair value as *"the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date"*. This definition makes it clear that fair value is an "exit price" from the perspective of the entity that holds the asset or has the liability.

Measurement of fair value

IFRS13 states that the fair values of assets and liabilities should be determined by using appropriate valuation techniques. Three techniques identified in IFRS13 are:

- (a) the "market approach", which uses prices generated by market transactions involving identical or comparable assets (e.g. quoted prices of listed shares)
- (b) the "cost approach", which is based upon current replacement costs
- (c) the "income approach", which converts future amounts (e.g. cash flows) into a single current discounted amount (e.g. present value).

IFRS13 also establishes a fair value hierarchy which categorises into three levels the inputs to the techniques used to measure fair value. These levels are as follows:

- (a) "Level 1 inputs" are quoted prices in active markets for identical assets or liabilities.
- (b) "Level 2 inputs" are inputs (other than Level 1 inputs) that are observable for the asset or liability, either directly or indirectly. Examples of such inputs include quoted prices for similar assets or liabilities in active markets and inputs other than quoted prices that are observable for the asset or liability, such as interest rates.
- (c) "Level 3 inputs" are unobservable inputs for the asset or liability (i.e. inputs for which market data is not available). Such inputs use the best information available about the assumptions that market participants would make when pricing the asset or liability.

IFRS13 gives the highest priority to Level 1 inputs and gives the lowest priority to Level 3 inputs. Entities are required to maximise the use of observable inputs and minimise the use of unobservable inputs when making fair value measurements.

Fair value disclosures

Entities are required to disclose information which helps users to assess:

- (a) for assets and liabilities measured at fair value in the entity's financial statements, the valuation techniques and inputs used to determine those measurements
- (b) for fair value measurements made using Level 3 inputs, the effect of those measurements on the entity's profit or loss or other comprehensive income for the period.

Exercises

A set of multiple choice questions for this chapter is available on the accompanying website.

- 5.1**
- (a) Define the term "property, plant and equipment".
 - (b) When should an item of property, plant and equipment be recognised as an asset and when should it be derecognised?
 - (c) Which of the following costs should be included in the cost of an item of property, plant and equipment at initial recognition?
 - the purchase price of the item
 - import duties incurred in relation to the item
 - delivery charges
 - small spare parts for the item
 - the cost of a maintenance contract
 - non-refundable purchase taxes relating to the purchase price of the item.
- 5.2**
- (a) Distinguish between the cost model and the revaluation model for the measurement of property, plant and equipment subsequent to its initial recognition.
 - (b) If the revaluation model is used, explain how revaluation gains and losses should be accounted for.
 - (c) A company which uses the revaluation model and prepares financial statements to 31 December each year acquired an item of property on 1 February 2019 at a cost of £5 million. On 31 December 2019, this property was revalued at £5.2 million. On 31 December 2020, it was revalued at £4.7 million.
Explain how these revaluations should be accounted for. Also explain what would have happened if the valuations at 31 December 2019 and 2020 had been reversed.
- 5.3**
- (a) In relation to property, plant and equipment, define the terms "depreciation", "depreciable amount", "useful life" and "residual value".
 - (b) On 1 July 2019, a company which prepares financial statements to 30 June each year acquires an item of equipment at a cost of £59,500. The item's useful life is expected to be five years with a residual value of £10,000. Calculate depreciation charges for each of the five years of the item's useful life, using:
 - the straight-line method
 - the diminishing balance method (at a rate of 30%)How should the company choose between these methods?

- 5.4** (a) Define the term "borrowing costs" and explain the accounting treatment of such costs which is required by international standard IAS23.
- (b) During the year to 31 December 2019, a company started work on the construction of a manufacturing plant and incurred expenditure as follows:

	<i>£000</i>
1 April 2019	1,500
1 August 2019	2,400
1 December 2019	1,800

All of these payments were made out of general borrowings. Construction work was still underway at 31 December 2019. The company had the following general borrowings outstanding throughout the year 2019:

	<i>Amount of loan</i>	<i>Interest for the year</i>
	<i>£000</i>	<i>£000</i>
Loan A	10,000	1,150
Loan B	8,000	720
Loan C	5,000	430

Calculate the amount of borrowing costs that should be capitalised in relation to the construction of the manufacturing plant during the year.

- 5.5** (a) Define the term "investment property" and explain the two models permitted by international standard IAS40 for the measurement of investment property after its initial recognition.
- (b) How do these two models differ from the two models permitted by IAS16 in relation to the measurement of property, plant and equipment?

- 5.6** Elite Leisure is a private limited liability company that operates a single cruise ship. The ship was acquired on 1 October 2011. Details of the cost of the ship's components and their estimated useful lives are:

component	original cost	depreciation basis
	(£million)	
ship's fabric (hull, decks etc.)	300	25 years straight-line
cabins and entertainment area fittings	150	12 years straight-line
propulsion system	100	useful life of 40,000 hours

At 30 September 2019 no further capital expenditure had been incurred on the ship.

In the year ended 30 September 2019 the ship had experienced a high level of engine trouble which had cost the company considerable lost revenue and compensation costs. The measured expired life of the propulsion system at 30 September 2019 was 30,000 hours. Due to the unreliability of the engines, a decision was taken in early October 2019 to replace the whole of the propulsion system at a cost of £140 million. The expected life of the new propulsion system was 50,000 hours and in the year ended 30 September 2020 the ship had used its engines for 5,000 hours.

At the same time as the propulsion system replacement, the company took the opportunity to do a limited upgrade to the cabin and entertainment facilities at a cost of £60 million and repaint the ship's fabric at a cost of £20 million. After the upgrade of the cabin and entertainment area fittings it was estimated that their remaining life was five years (from the date of the upgrade). For the purpose of calculating depreciation, all the work on the ship can be assumed to have been completed on 1 October 2019. All residual values can be taken as nil.

Required:

Calculate the carrying amount of Elite Leisure's cruise ship at 30 September 2020 and the related amounts which should be recognised as expenses for the year ended 30 September 2020. Your answer should explain the treatment of each item.

(ACCA)

- *5.7** Accounting standard IAS16 *Property, Plant and Equipment* makes a number of recognition, measurement and disclosure requirements with regard to tangible non-current assets. The term "non-current asset" is defined in accounting standard IAS1 *Presentation of Financial Statements*. The information given below relates to two companies, both of which prepare accounts to 31 December.

Alpha Ltd

Alpha Ltd bought a factory machine on 30 June 2019 and paid a total of £420,000. The supplier's invoice showed that this sum was made up of the following items:

	£
Manufacturer's list price	380,000
<u>Less: Trade discount</u>	<u>38,000</u>
	342,000
Delivery charge	6,800
Installation costs	29,600
Maintenance charge for year to 30 June 2020	27,000
Small spare parts	<u>14,600</u>
	<u>420,000</u>

Beta Ltd

On 1 January 2009, Beta Ltd bought freehold property for £800,000. This figure was made up of land £300,000 and buildings £500,000. The land was non-depreciable but it was decided to depreciate the buildings on the straight-line basis, assuming a useful life of 40 years and a residual value of £nil.

On 1 January 2019, the land was revalued at £400,000 and the buildings were revalued at £450,000. The company decided to incorporate these valuations into its accounts. The previous estimates of the buildings' useful life and residual value remain unchanged.

Required:

- (a) Distinguish between current assets and non-current assets, giving TWO examples of each type of asset.
- (b) Explain, with appropriate examples, the difference between capital expenditure and revenue expenditure.
- (c) If a company incorrectly classified an item of capital expenditure as revenue expenditure, what effect would this have on the company's accounts in the year of the expenditure and in subsequent years?
- (d) In accordance with the rules of IAS16, calculate the cost figure at which the machine bought by Alpha Ltd should initially be measured. Also explain the correct accounting treatment of any component of the £420,000 expenditure which cannot be treated as part of the machine's cost.
- (e) Write journal entries for the revaluation of Beta Ltd's freehold property on 1 January 2019. Also calculate the amount of depreciation which should be charged in relation to the buildings for the year to 31 December 2019.

*(CIPFA)****5.8**

- (a) Define the term "investment property". Explain why it is not generally appropriate to charge depreciation in relation to such a property.
- (b) Give three examples of properties (land or buildings) that should be classified as investment properties.
- (c) Explain the "fair value model" which is permitted by IAS40 for the measurement of investment property.
- (d) A company's investment property at 31 December 2019 is shown in the statement of financial position at its fair value of £23m. Explain the accounting treatment which should be applied in the financial statements for the year to 31 December 2020 if the property has risen in value by 10% during the year. Also explain the accounting treatment if the property has fallen in value by 10% during the year.